BACSWN Edge AI Node Hardware Budget

BACSWN — Edge AI Agent Node: Hardware Budget (Towers)

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1. Purpose & Scope

This is a **budgetary hardware estimate** for deploying the autonomous **AI agent node** at each of the 15 tower/weather stations — the compute, connectivity, sensing, power, and protection that turn a tower into an intelligent BACSWN station. Figures are **indicative planning estimates (USD), not vendor quotes**, intended to size the hardware program; firm pricing requires RFQs.

In scope: per-node edge hardware (compute, radios, sensors, power, enclosure/protection); the **central hub / NOC and off-island standby room infrastructure** (racks, servers, network, UPS, cooling, **fire suppression**, physical security, NOC fit-out); spares, assembly/integration & factory test, depot tooling, contingency. **Out of scope (budgeted elsewhere):** tower/mast civil structures and foundations, the **Raytheon primary radar** program (WS-P), and all **recurring costs** (Iridium airtime, cellular plans, power, licences) which are operating expenses, not hardware.

2. Key Finding (read this first)

The AI itself is still the cheap part. Even with a **multimodal model** and a **2-of-3 model-consensus ensemble** (§7), the edge-AI compute is only **~$2,100/node** — about **4%** of an instrumented node. The cost is dominated by **aviation-grade meteorological sensors** (especially the ceilometer and visibility sensor) and **resilient solar/battery power**, and now also the **central hub/NOC room** (cooling, UPS, fire suppression — §5). Budget decisions should focus on the *sensor-suite tier*, *power autonomy*, and *facility*, not the AI compute.

3. Assumptions & Basis

- 15 stations; one agent node each. Costs are mid-points within the ranges shown.
- Aviation-grade (AWOS/ASOS-class) sensors assumed at all sites in the baseline (see §6 for a tiered alternative that materially reduces cost).
- VHF baseline uses the architecture's **Motorola MTR3000** repeater-grade radio; §6 notes a data-radio alternative.
- Remote/relay nodes (MYBC, MYIG, MYMM, MYRD) carry larger batteries (96–120 h) — a per-node uplift.
- Excludes shipping/duties, install labour (civil), and recurring airtime/licences.
- **Trusted supply chain:** edge compute is NVIDIA (US-built); AI **models must be non-Chinese, trusted-provenance** (policy: Data Governance BACSWN-GOV-2026-005) — a procurement constraint on the model ensemble, not a hardware cost.

4. Per-Node Bill of Materials (indicative)

Tier	Item	Unit cost (USD, range)	Midpoint
A — Edge AI compute	NVIDIA Jetson Orin NX 16GB (~100 TOPS) + industrial carrier — runs the **multimodal 2-of-3 model ensemble**	$1,300–1,900	$1,600
	Ruggedized compute enclosure, cooling, conformal coat	$250–450	$300
	Industrial storage (SSD — larger for models/imagery)	$120–200	$150
	Secure element / TPM 2.0 (FIPS)	$20–60	$40
	Subtotal — AI compute		**~$2,090**
B — Connectivity	VHF mesh radio (MTR3000) + duplexer + antenna/feeder	$4,000–6,000	$5,000
	Iridium 9603 SBD transceiver + antenna	$500–700	$600
	Industrial LTE modem + antenna	$300–450	$400

Tier	Item	Unit cost (USD, range)	Midpoint
	Subtotal — connectivity		**~$6,000**
C — Meteorological sensors	Temp / humidity / pressure	$700–1,200	$1,000
	Ultrasonic anemometer (wind)	$1,200–1,800	$1,500
	Visibility / present-weather sensor	$6,000–10,000	$8,000
	Ceilometer (cloud base, laser)	$18,000–30,000	$22,000
	Precipitation + lightning sensor	$1,200–2,000	$1,500
	Subtotal — sensors		**~$34,000**
D — Power & protection	Solar array (48 V) + MPPT charge controller	$2,000–3,000	$2,500
	Battery bank (72–120 h autonomy)	$3,000–4,500	$3,500
	Grid-tie + automatic transfer switch	$600–1,000	$800
	NEMA-4X / IP66 missile-rated enclosures	$1,200–1,800	$1,500
	Lightning protection, SPDs, grounding	$900–1,500	$1,200
	Subtotal — power & protection		**~$9,500**
	PER-NODE TOTAL (baseline)		**~$51,600**

5. Facility & Room Infrastructure (Central Hub / NOC + Standby)

The towers are self-contained outdoor enclosures, but the **central hub / Network Operations Centre** and the **off-island standby** need a proper equipment room. *(A cloud region may serve as the standby instead — that shifts most of §5.2 to operating cost.)*

5.1 Primary Hub / NOC (Nassau)

Item	Indicative (USD)
Server racks / cabinets + PDUs (≈4 racks)	$12,000
Hub server cluster (control plane, COP, database; redundant) + storage	$60,000
Network — redundant switches, routers, firewalls	$25,000
UPS (N+1, room-level)	$30,000
Standby generator (or building tie-in)	$40,000
Precision cooling / CRAC (N+1)	$40,000
Fire detection + clean-agent suppression (Novec 1230 / FM-200)	$35,000
Raised floor / room build-out / aisle containment	$30,000
Physical security (access control, CCTV, intrusion)	$20,000
Structured cabling, PDUs, KVM	$15,000
Environmental monitoring (temp / humidity / leak / smoke)	$8,000
NOC operations centre — operator workstations + video wall / displays	$40,000
Subtotal — primary hub / NOC	**~$355,000**

5.2 Off-Island Standby (DR)

Item	Indicative (USD)
Lighter room — racks, standby servers, network, UPS, cooling, basic fire suppression, security	$120,000
Cloud-DR alternative shifts most of this to opex	*~$15,000 capex*
Subtotal — standby	**~120,000(*or* 15,000 cloud)**

Facility subtotal: ~$475,000 (≈ $370,000 with a cloud standby).

6. Network & Facility Roll-Up

Line	Basis	Amount (USD)
Agent nodes — 15 × ~$51,300	Baseline per-node	~$770,000
Remote-node battery uplift (4 sites, extended autonomy)	+~$2,000 ×4	~$8,000
Facility & room infrastructure (hub/NOC + standby)	§5	~$475,000
Spares pool	~15% of nodes	~$115,000
Assembly, integration & factory acceptance test	labour/build	~$120,000
Depot / field test equipment & tooling	one-time	~$50,000
Subtotal		**~$1,538,000**
Contingency	~15%	~$230,000
HARDWARE PROGRAM TOTAL (baseline)		**~$1.77M**

*Indicative range overall: **~$1.6M–2.2M** depending on sensor tier, VHF radio choice, and cloud-vs-physical standby.*

7. Cost Drivers & Options

- **Sensor tiering (largest lever).** The ceilometer + visibility sensor are **~60%** of node cost. A **tiered suite** — full aviation instrumentation at major/hub sites, a lighter suite (no ceilometer) at selected remote sites — could cut the network hardware total by **~$300,000–450,000**. This is a meteorological-coverage decision for the watch office, not just a budget one.
- **VHF radio.** Substituting a base/mobile **data radio** for a full MTR3000 **repeater** at non-relay nodes could save **~$2,500–3,500/node**; keep repeater-grade only at relay hubs (e.g., Chub Cay).
- **AI compute & model consensus.** The tower model must be **multimodal** (sensor time-series + radar/satellite imagery + METAR/text) and run a **2-of-3 model-consensus ensemble** — three independent models must reach majority before an autonomous correction/action. The baseline budgets one capable **Orin NX 16GB** running the ensemble. For higher assurance, **three independent compute modules** (true hardware redundancy) add ~$3,000–4,000/node; **AGX Orin** is an option at hub/relay nodes. Still modest against sensors/power.
- **Volume procurement.** A single 15-node (+ spares) buy should attract meaningful vendor discounts not reflected in these list-level estimates.

8. Exclusions (budgeted separately)

Tower/mast structures and foundations (WS-G civil); the **Raytheon primary radar** program (WS-P); IoT sensor network and drones (separate BOMs); and all **recurring** costs — Iridium airtime, cellular plans, spectrum/licences, and O&M — which are operating expenses, not node hardware.

Indicative budgetary estimates for planning only; not a financial model and not vendor-quoted. Confirm with RFQs before commitment.